# Tarek Messaoudi

Full Stack Engineer, Medical Imaging

tarek.messaoudi.dev | github.com/tmessaoudi | tarek@messaoudi.dev | /tmessaoudi

## Work Experience

**Software Engineer** — **2026 - Present**

**Chipiron** // An MRI scanner company

🖥️ MRI Operator Web Application

- Built the app from the first commit as its main developer, shipping 14 feature releases in six months
- Designed the exam workflow: drag-and-drop protocols, step-based execution, live scan progress, abort
- Built a JSON Schema-driven form engine for acquisition parameters, with server-side field validation

🩻 DICOM Imaging Integration

- Connected Orthanc PACS storage, an authenticated DICOMweb proxy and a Cornerstone.js viewer

🔐 Access Control & Quality

- Implemented authentication and role-based access control in a FastAPI service
- Introduced front-end testing from scratch (Vitest, MSW, Playwright) and OpenAPI-generated API types

**Senior Full Stack Engineer** — **2020 - 2025**

**Qynapse** // A healthcare technology company specializing in neuroimaging and AI

🧠 Medical Imaging Web Platform Redesign

- Refactored with a new admin panel, interactive data visualizations, and a robust MRI viewer, ...
- Added segmentation correction, lesion tracking, 3D rendering, oblique views and multi-modality

📝 Reporting System Overhaul

- Rebuilt the report pipeline (Jinja2 + Puppeteer) with templates per language and disease (Dementia, MS, ...)

📦 Scalable Infrastructure & Microservices

- Contributed to converting the academic BrainVisa platform into a Kubernetes-based scalable pipeline
- Expanded the customer-deployed DIMSE agent with routing logic and detailed logging

🧬 Domain-Specific Tools

- Built a normative Python library for imaging metrics (z-scores, percentiles, asymmetry, ...)
- Developed a 3D mesh microservice for structural MRI data

**Full Stack Engineer** — **2017 - 2020**

**Alter Solutions** // A consulting and IT services company

- Built web apps from scratch for IRT System X (research dashboard), Thalès (inventory), PSA (file parser)
- Refactored SFR Altice's customer-service CRM with a new architecture and Mail/SMS notifications

## Technical Skills

**Programming Languages:** TypeScript, JavaScript, Python, Java
**Frontend Development:** React, Tailwind CSS, Remix
**Backend Development:** NodeJs, Deno, FastApi, Spring
**Databases:** PostgreSQL, SQLite, MongoDB, REDIS
**Other Technologies:** Docker, Kubernetes, Git, Linux

## Education

**IMT Lille-Douai** — **2012 - 2017**

**Software engineer major** // with a foreign semester in South Korea at Chungnam National Univ.

## Projects & Rewards

**LOM** (GitHub) - *Browser strategy game*

**"Entreprise du Futur" Reward** (Site) - *By Thalès, from Challenge Entreprendre*